## Key Assessment – Final Project (40% of Overall Grade)

| Criteria [with Weight] | Above Standard (Broad wealth of knowledge, excellent presentation) (100%) | At Standard (Accurate, general knowledge, well presented) (81-90%) | Approaching Standard (Moderate knowledge, adequately expressed) (71-80%) | Below Standard (Limited conceptual understanding, inadequately expressed) (70% or below) | Row Score |
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| **Project Topic, Existing Research Review and Problem Definition. [20]** | Research is thorough, the problem is well-defined. Literature review is done. Appropriate and accurate information that provides the chosen topic's machine learning/artificial intelligence capability. Minimums were met. | Research includes appropriate and accurate information about the chosen topic's Machine-learning/artificial intelligence capability. Minimums were met. | In parts, research includes appropriate and accurate information about the chosen topic's machine learning/artificial intelligence capability. Minimums were or were not met. | Research is either inappropriate or contains inaccurate information about the topic. Minimums were not met. | |
| **Presentation Quality and End Result [20]** | The presentation was fully completed, and the media (text, images etc.) was very well designed for presentation. Results provide excellent insights into the solution for the problem at hand. Final paper/report is ready to be posted to arXiv as a research paper. | The presentation was completed, and the media was well-designed for presentation. Results provide good insights to the solution and the problem. Final paper/report presents the analysis well. | At least half of the presentation was fully completed, and the media was somewhat designed for presentation. Results partially provide insights to the solution and problem. Final paper/report approaches standard. | Less than half of the presentation was completed, and the media was not designed for presentation. Results do not provide any insight into the solution or the problem. Final paper/report inadequate. | |
| **Problem Solving Techniques/Models used and Implemented [40]** | Choice of models and other techniques used to solve the problem are justified by extensively using Machine Learning/AI theory. Implementation(code) exhibits excellent understanding of ML coding techniques and knowledge of packages and frameworks. Code is reproducible for the problem solved. Code is well documented on GitHub. | Choice of models and other techniques used to solve the problem are justified (ML theory is not necessarily used). Implementation (code) exhibits good understanding of ML coding techniques and knowledge of packages and frameworks. | Choice of models and other techniques used to solve the problem are somewhat justified (ML theory is not necessarily used). Implementation has minor issues. | No justification for the choice of models or techniques used. Implementation has errors. No documentation of code. | |

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| **Data Exploration and Usage [20]** | More than 2 datasets are used for analysis. Extensive research is done to find appropriate datasets/data sources. Detailed experimentation done to understand the data. Pre-processing is justified with respect to the problem at hand. Extensive experimentation was done on data preprocessing. | 2 datasets used for analysis. Appropriate research is done to find appropriate datasets/data sources. Good experimentation done to understand the data. Pre-processing justified with respect to the problem at hand. | Only 1 dataset is used. Data choice, exploration and data pre-processing done. | Not enough data exploration and/or pre-processing. | |
| **Total** | | | | | /100 |

**Note:** *The above rubric is designed for the Final Project grading. Total points are out of 100. The elements in the first column are 4 major criteria used for grading. The weights are given in the first row.*